# Basic Arithmetic Operations

## Chapter 4: HCF and LCM

---

1. Find the LCM of 8, 15, 24 and 72.

   a) 3500 b) 360 c) 720 d) 735 e) None of the above

2. If three numbers are $2a, 5a$ and $7a$, what will be their LCM?

   a) $70a$ b) $65a$ c) $75a$ d) $70a^3$ e) None of the above

3. Find the LCM of $(2^3 \times 3 \times 5^2 \times 7)(2^4 \times 3^2 \times 5 \times 7^2 \times 11)$ and $(2 \times 3^3 \times 5^4)$

   a) $2^4 \times 3^3 \times 5^4$ b) $2 \times 3 \times 7 \times 5 \times 11$ c) $2^4 \times 3^3 \times 5^4 \times 7^2 \times 11$

   d) $2^4 \times 3^4 \times 5^4 \times 7$ e) None of the above

4. Find the HCF of 132, 204 and 228.

   a) 12 b) 18 c) 6 d) 21 e) None of the above

5. What will be the HCF of $(2 \times 3 \times 7 \times 9)$, $(2 \times 3 \times 9 \times 11)$ and $(2 \times 3 \times 4 \times 5)$?

   a) $2 \times 3 \times 7$ b) $2 \times 3 \times 9$

   c) $2 \times 3$ d) $2 \times 7 \times 9 \times 11$

6. Find the LCM of $\frac{1}{3}, \frac{2}{9}, \frac{5}{6}, \frac{4}{27}$.

   a) $\frac{1}{54}$ b) $\frac{10}{27}$ c) $\frac{20}{3}$ d) $\frac{3}{20}$

7. Find the LCM of $\frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{9}{13}$

   a) 36 b) $\frac{1}{36}$ c) $\frac{1}{1365}$ d) $\frac{12}{455}$

8. Find the HCF of $\frac{4}{5}$ and $\frac{7}{15}$

   a) $\frac{1}{13}$ b) $\frac{1}{5}$ c) $\frac{1}{15}$ d) $\frac{1}{25}$ e) None of the above

9. Find the HCF of $\frac{1}{2}, \frac{3}{4}, \frac{4}{5}$

   a) $\frac{1}{20}$ b) $\frac{1}{40}$ c) 20 d) 15 e) None of the above

10. Which of the following will be the LCM of 0.25, 0.1 and 0.125?

    a) 0.25 b) 0.005 c) 0.05 d) 0.5 e) None of the above

11. Find the LCM of 2.5, 1.2, 20 and 7.5.

    a) 60 b) 65 c) 70 d) 50 e) None of the above

12. Product of two coprime numbers is 117. Then, their LCM is

    a) 9 b) 13 c) 39 d) 117

13. The product of HCF and LCM of 18 and 15 is

    a) 120 b) 150 c) 175 d) 270

14. The LCM of two numbers is 2376 while their HCF is 33. If one of the number is 297, then the other number is

    a) 216 b) 264 c) 642 d) 792

15. The HCF and LCM of two numbers are 13 and 1989, respectively. If one of the numbers is 117, then determine the other

    a) 121 b) 131 c) 221 d) 231

16. The HCF of two numbers is 15 and their LCM is 225. If one of the numbers is 75, then find the another number.

    a) 105 b) 90 c) 60 d) 45

17. If HCF of two numbers is 8, which of the following can never be their LCM?

    a) 24 b) 48 c) 56 d) 60 e) None of the above

18. The difference of two numbers is $\frac{1}{9}$ of their sum. Their sum is 45. Find the LCM.

    a) 225 b) 100 c) 150 d) 200 e) None of the above

19. The ratio of two numbers is 3 : 4 and their HCF is 4. What will be their LCM?

    a) 12 b) 16 c) 24 d) 48 e) None of the above

20. The ratio of two numbers is 5 : 6 and their LCM is 480, then their HCF is

    a) 20 b) 16 c) 6 d) 5

21. The HCF of three numbers is 23. If they are in the ratio of 1: 2: 3, then find the numbers.

    a) 69, 15, 22 b) 23, 46, 69 c) 25, 31, 41 d) 23, 21, 35 e) None of the above

22. Three numbers are in the ratio of 3 : 4 : 5 and their LCM is 1200. Find the HCF of the numbers.

    a) 40 b) 30 c) 80 d) 20 e) None of the above

23. The HCF and LCM of two numbers $m$ and $n$ are respectively 6 and 210. If $m + n = 72$, then $\frac{1}{m} + \frac{1}{n}$ is equal to

    a) $\frac{1}{35}$ b) $\frac{3}{35}$ c) $\frac{5}{37}$ d) $\frac{2}{35}$ e) None of the above

24. If a number is exactly divisible by 11 and 13, which of the following types the number must be?

    a) Divisible by (11 + 13) b) Divisible by (13-11) c) Divisible by $(11 \times 13)$

    d) Divisible by $(13 \div 11)$ e) None of the above

25. The LCM of two numbers is 48. The numbers are in the ratio of 2 : 3. Find the sum of the numbers.

    a) 28 b) 32 c) 40 d) 64

26. Four numbers are in the ratio of 10 : 12 : 15 : 18. If their HCF is 3, then find their LCM.

    a) 420 b) 540 c) 620 d) 680 e) None of the above

27. The product of two whole numbers is 1500 and their HCF is 10. Find the LCM.

    a) 15000 b) 150 c) 1500 d) 15 e) None of the above

28. If the HCF of $a$ and $b$ are 12 where $a, b$ are positive integers with $a > b > 12$, then what will be the values of $a$ and $b$?

    a) 12,24 b) 24,12 c) 24,36 d) 36,24

29. The sum of HCF and LCM of two numbers is 403 and their LCM is 12 times their HCF. If one number is 93, then find the another number.

    a) 115 b) 122 c) 124 d) 138 e) None of the above

30. The LCM of two numbers is 495 and their HCF is 5. If sum of the numbers is 100, then find the difference of the numbers.

    a) 10 b) 46 c) 70 d) 90 e) None of the above

31. The LCM of two numbers is 20 times of their HCF and (LCM + HCF) = 2520. If one number is 480, what will be the triple of another number?

    a) 1200 b) 1500 c) 2100 d) 1800 e) None of the above

32. The sum of two numbers is 1056 and their HCF is 66, find the number of such pairs.

    a) 6 b) 2 c) 4 d) 8 e) None of the above

33. What is the smallest possible length that can be exactly measured by the scales of lengths 3 cm, 5 cm and 10 cm?

    a) 15 cm b) 30 cm c) 28 cm d) 40 cm e) None of the above

34. What is the least number which is exactly divisible by 8, 9, 12, 15 and 18 and is also a perfect square?

    a) 3600 b) 7200 c) 5200 d) 6500 e) None of the above

35. Find the greatest number of 3-digits which when divided by 6, 9, 12 leaves 3 as remainder in each case.

    a) 975 b) 996 c) 903 d) 939 e) None of the above

36. What will be the greatest number that divides 1356, 1868 and 2764 leaving 12 as remainder in each case?

    a) 64 b) 124 c) 156 d) 260

37. Find the greatest number that divides 130, 305 and 245 leaving remainders 6, 9 and 17, respectively?

    a) 4 b) 5 c) 14 d) 24 e) None of the above

38. What will be the greatest number that divides 1023 and 750 leaving remainders 3 and 2, respectively?

    a) 68 b) 65 c) 78 d) 19 e) None of the above

39. In a store, there are 345 *l* mustard oil, 120 *l* sunflower oil and 225 *l* soyabean oil. What will be the capacity of the largest container to measure the above three types of oil?

    a) 8 *l* b) 20 *l* c) 23 *l* d) 15 *l* e) None of the above

40. The least number which should be added to 2497, so that the sum is exactly divisible by 5, 6, 4 and 3, is

    a) 3 b) 13 c) 23 d) 33 e) None of the above

41. Find the largest number which divides 1305, 4665 and 6905 leaving same remainder in each case. Also, find the common remainder.

    a) 1210, 158 b) 1120, 158 c) 1120, 185 d) 1210, 185 e) None of the above

42. The least number which when divided by 12, 16 and 18 leaves 5 as remainder in each case. Find the number.

    a) 139 b) 144 c) 149 d) 154 e) None of the above

43. What is the greatest number that divides 13850 and 17030 leaves a remainder 17?

    a) 477 b) 159 c) 107 d) 87

44. The HCF and LCM of two natural numbers are 12 and 72, respectively. What is the difference between the two numbers, if one of the number is 24?

    a) 12 b) 18 c) 21 d) 24

45. What will be the least number which when divided by 12, 21 and 35 leaves 6 as remainder in each case?

    a) 426 b) 326 c) 536 d) 436

46. What is the least number which when diminished by 7, is divisible by each one of 21, 28, 36 and 45?

    a) 1255 b) 1177 c) 1265 d) 1267 e) None of the above

47. What is the least number which when increased by 9, is divisible by each one of 24, 32, 36 and 54?

    a) 855 b) 890 c) 756 d) 895 e) None of the above

48. Find the least number which when divided by 16, 18 and 20 leaves a remainder 4 in each case, but is completely divisible by 7.

    a) 2884 b) 2256 c) 865 d) 3332 e) None of the above

49. What is the greatest four digit number which when divided by 10, 15, 21 and 28 leaves remainders 4, 9, 15 and 22, respectively?

    a) 9654 b) 9666 c) 9664 d) 9864 e) None of the above

50. How many numbers are there between 4000 and 6000 which are exactly divisible by 32, 40, 48 and 60?

    a) 2 b) 3 c) 4 d) 5

51. Three pieces of timber 84 $m$, 98 $m$ and 126 $m$ long have to be divided into planks of the same length. What is the greatest possible length of each plank?

    a) 14 $m$ b) 28 $m$ c) 7 $m$ d) 21 $m$ e) None of the above

52. In a school, all the students can stand in a row, so that each row has 5, 9 or 10 students. Find the least number of students in the school.

    a) 90 b) 95 c) 85 d) 100

**Exercise : Higher Skill Level Questions**

1. If $(x-6)$ is the HCF of $x^2-2x-24$ and $x^2-kx-6$, then what is the value of $k$

    a) 3 b) 5 c) 6 d) 8

2. If $a$ and $b$ are positive integers, then what is the value of

$$\left(\frac{a}{\text{HCF of } (a,b)}, \frac{b}{\text{HCF of } (a,b)}\right)$$

    a) $a$ b) $b$ c) 1 d) $\frac{a}{\text{HCF of } (a,b)}$

3. What is the HCF of $8(x^5 - x^3 + x)$ and $28(x^6 + 1)$?

   a) $4(x^4 - x^2 + 1)$ b) $(x^3 - x + 4x^2)$ c) $(x^3 - x + 3x^2)$ d) None of the above

4. The HCF of $(x^4 - y^4)$ and $(x^6 - y^6)$ is

   a) $x^2 - y^2$ b) $x - y$ c) $x^3 - y^3$ d) $x^4 - y^4$

5. The HCF of $(x^3 - y^2 - 2x)$ and $(x^3 + x^2)$ is

   a) $x^3 - x^2 - 2x$ b) $x^2 + x$ c) $x^4 - x^3 - 2x^2$ d) $x - 2$

6. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$

   a) $ab$ b) $a^2b^3$ c) $a^2b^2$ d) $a^3b^2$

7. For any integer $n$, what is HCF $(22n + 7, 33n + 10)$ equal to?

   a) 0 b) 1 c) 11 d) None of these

8. For any integers $a$ and $b$ with HCF $(a, b) = 1$; what is HCF $(a + b, a - b)$ equal to?

   a) It is always 1 b) It is always 2 c) Either 1 or 2 d) None of the above

9. In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooter hit the target once in every 5 $s$, 6$s$, 7$s$ and 8 $s$ respectively. If all of them hit their targets at 9:00 am, when will they hit their targets together again?

   a) 9 : 04 am b) 9 : 08 am c) 9 : 14 am d) None of the above

10. Three bells chime at intervals of 48, 60 and 90 min, respectively. If all the three bells chime together at 10:00 am, at what time will all the three chime again that day?

    a) 1:00 pm b) 2:00 pm c) 8:00 pm d) 10:00 pm

11. Five bells begin to toll together at intervals of 9 $s$, 6$s$, 4$s$, 10$s$ and 8$s$ respectively. How many times will they toll together in the span of 1$h$ (excluding the toll at the start)?

    a) 5 b) 8 c) 10 d) Couldn't be determined e) None of the above

12. Monica, Veronica and Rachat begin to jog around a circular stadium. They complete their revolutions in 42$s$, 56$s$ and 63$s$, respectively. After how many seconds will they be together at the starting point?

    a) 366 b) 252 c) 504 d) Couldn't be determined e) None of the above

13. A General can draw up his soldiers in the rows of 10, 15 or 18 soldiers and he can also draw them up in the form of a solid square. Find the least number of soldiers with the General.

    a) 100 b) 3600 c) 900 d) 90

14. 6 bells commence tolling together and toll of intervals are 2, 4, 6, 8, 10 and 12 seconds, respectively. In 1 h, how many times, do they toll together?

    a) 16 b) 32 c) 21 d) 35 e) None of the above

15. Find the greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm and 12 m 95 cm.

    a) 15 cm  b) 25 cm  c) 35 cm  d) 42 cm  e) 45 cm

16. The HCF and LCM of two numbers are 21 and 4641, respectively. If one of the numbers lies between 200 and 300, then find the two numbers.

    a) 273, 363  b) 273, 359  c) 273, 361  d) 273, 357  e) None of the above

17. Find the side of the largest possible square slabs which can be paved on the floor of a room 2 m 50 cm long and 1 m 50 cm broad. Also find the number of such slabs to pave the floor.

    a) 25, 20  b) 30, 15  c) 50, 15  d) 55, 10  e) None of the above

18. When in each box 5 or 6 dozens of apples were packed, three dozens were left. Therefore, bigger boxes were taken to pack 8 or 9 dozens of apples. However, still three dozens of apples remained. What was the least number of dozens of apples to be packed?

    a) 363 dozens  b) 315 dozens  c) 345 dozens  d) 335 dozens  e) None of the above

**Basic Skill Level Questions Answer Key**

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | b | 6 | c | 11 | a | 16 | d | 21 | b | 26 | b | 31 | d | 36 | a | 41 | c | 46 | d | 51 | a |
| 2 | a | 7 | a | 12 | d | 17 | d | 22 | d | 27 | b | 32 | c | 37 | a | 42 | c | 47 | a | 52 | a |
| 3 | c | 8 | c | 13 | d | 18 | b | 23 | d | 28 | d | 33 | b | 38 | a | 43 | b | 48 | a | | |
| 4 | a | 9 | a | 14 | b | 19 | d | 24 | c | 29 | c | 34 | a | 39 | d | 44 | a | 49 | a | | |
| 5 | c | 10 | e | 15 | c | 20 | b | 25 | c | 30 | a | 35 | a | 40 | c | 45 | a | 50 | c | | |

**Higher Skill Level Questions**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | b | 6 | c | 11 | c | 16 | d |
| 2 | c | 7 | b | 12 | c | 17 | c |
| 3 | a | 8 | c | 13 | c | 18 | a |
| 4 | a | 9 | c | 14 | e | | |
| 5 | c | 10 | d | 15 | c | | |

# Chapter 5: Square Root and Cube Root

1. Find the square root of 144.

   a) 15 b) 16 c) 12 d) 13 e) None of the above

2. What will be the square root of 792

   a) 89 b) 87 c) 37 d) 47

3. The square root of $0.\overline{4}$ is

   a) $0.\overline{6}$ b) $0.\overline{7}$ c) $0.\overline{8}$ d) $0.\overline{9}$ e) None of the above

4. Find the squareroot of $105\frac{4}{64}$.

   a) $15\frac{1}{4}$ b) $15\frac{12}{4}$ c) $10\frac{1}{4}$ d) $6\frac{2}{4}$ e) None of the above

5. $\sqrt{\frac{324}{81}} + \sqrt{\frac{324}{81}} = ?$

   a) 4 b) 12 c) 6 d) 18 e) None of the above

6. The square root of $\left(\frac{1}{4}\right) \times \left(\frac{1}{49}\right) \div \left(\frac{25}{121}\right)$ is

   a) $\frac{11}{5}$ b) $\frac{11}{70}$ c) $\frac{7}{11}$ d) $\frac{11}{7}$ e) None of the above

7. Which of the following is false?

   a) $\sqrt{5184}=72$ b) $\sqrt{15625}=125$ c) $\sqrt{1444}=38$ d) $\sqrt{1296}=34$ e) None of the above

8. $\frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = ?$

   a) $2\sqrt{6}$ b) 2 c) $6\sqrt{2}$ d) $\frac{2}{\sqrt{6}}$ e) None of the above

9. $\sqrt{1089} + \sqrt{289} = \sqrt{?}$

   a) 625 b) 50 c) 2500 d) 1378 e) None of the above

10. $\sqrt{100\frac{1}{4}} = ?$

    a) 19.5 b) 10.25 c) 10.5 d) 11.5 e) None of the above

11. If $\sqrt{\frac{x}{169}} = \frac{18}{13}$, then $x$ is equal to

a) 106 b) 324 c) 2916 d) 4800 e) None of the above

12. The value of $\sqrt[6]{0.000729}$ is

a) 0.027 b) 0.3 c) 0.03 d) 0.09 e) None of the above

13. Evaluate $\sqrt{16} + \sqrt{\frac{9.5 \times 0.0085 \times 18.9}{0.021 \times 0.0017 \times 1.9}}$.

a) 154 b) 158 c) 160 d) 169 e) None of the above

14. Evaluate $\sqrt{\frac{0.289}{0.00121}} + \sqrt{\frac{64}{16}}$

a) $\frac{192}{11}$ b) $\frac{170}{11}$ c) $\frac{182}{11}$ d) $\frac{172}{11}$ e) None of the above

15. $\sqrt{15 \times 163 \div 5 - 89}$=?

a) 15 b) 25 c) 10 d) 20 e) None of the above

16. Find the square root of $\frac{0.204 \times 42}{0.07 \times 3.4}$

a) 5 b) 6 c) 3 d) 9 e) None of the above

17. $\sqrt{15^2 + 11 \times 3^2}$ =?

a) 18.25 b) 19 c) 18 d) 19.5 e) None of the above

18. The value of $\sqrt{400} + \sqrt{0.04} - \sqrt{0.000004}$ i

a) 20.22 b) 20.198 c) 20.188 d) 20.022 e) None of the above

19. $\sqrt{176 + \sqrt{2401}}$ =?

a) 14 b) 15 c) 18 d) 24 e) None of the above

20. $\sqrt{?} + 136 = \frac{5}{8}$ of 320

a) 1936 b) 4624 c) 4196 d) 4096 e) None of the above

21. $\frac{2707}{\sqrt{?}} = 27.07$

a) 10 b) 100 c) 1000 d) 10000 e) None of the above

22. $(64)^2 \div \sqrt[3]{32768}$ =?

a) 128 b) 132 c) 142 d) 104 e) None of the above

23. $\sqrt{?} + \frac{3}{5}$ of $80 = 60 \times \frac{1}{2} \times 8$

a) 36864 b) 46864 c) 56864 d) 66864 e) None of the above

24. $140\sqrt{?} + 315 = 1015$

a) 16 b) 25 c) 36 d) 5 e) None of the above

25. If $\sqrt{3} = 1.732$, then find the value of $4(\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75})$ correct to 3 places of decimal

a) 6.928 b) 5.928 c) 2.732 d) 3.732 e) None of the above

26. If $\frac{\sqrt{1296}}{x} = \frac{x}{2.25}$,then find the value of $x$.

a) 6 b) 7 c) 8 d) 9 e) None of the above

27. If $\sqrt{1 + \frac{x}{144}} = \frac{13}{12}$, then find the value of $x$

a) 1 b) 12 c) 13 d) 25 e) None of the above

28. If $x = 2 + \sqrt{2}$ and $y = 2 - \sqrt{2}$, then find the value of $(x^2 + y^2)$

a) 12 b) 14 c) 6 d) 18 e) None of the above

29. $(\sqrt{65025})^2 = (?)^2$

a) 65.025 b) 32512.5 c) 255 d) 510 e) None of the above

30. $\sqrt{?} + 43 = \sqrt{19881}$

a) 9604 b) 7744 c) 9216 d) 8464 e) None of the above

31. $\sqrt{8 \times 7 - ? + 208 \div 16} = 18$

a) 5 b) 7 c) 4 d) 3 e) None of the above

32. If $0.14 \div x^2 = 14$, then $x$ is equal to

a) 0.01 b) 10 c) 100 d) 0.1 e) None of the above

33. If $\sqrt{18 \times 14 \times x} = 168$, then $x$ is equal to

a) 113 b) 112 c) 115 d) 117 e) None of the above

34. $22^2 + \sqrt{?} = 516$

a) 1028 b) 1024 c) 1124 d) 1128 e) None of the above

35. $\sqrt{?} + 28 = \sqrt{1681}$

a) 13 b) 225 c) 216 d) 169 e) None of the above

36. Evaluate $\sqrt{0.9}$ upto 3 places of decimal

a) 0.948 b) 0.984 c) 0.988 d) 0.938 e) None of the above

37. Sukhiram plants 15376 orange trees in his garden and arranges them, so that there are as many rows as there are orange trees in each row. Find the number of rows.

a) 125 b) 124 c) 128 d) 135 e) None of the above

38. What is the least number to be multiplied with 294 to make it a perfect square?

a) 2 b) 3 c) 6 d) 24 e) None of the above

39. What least number should be subtracted from 6860, so that 19 be the cube root of the result from this subtraction?

a) 1 b) 2 c) 3 d) 5 e) None of the above

40. What is the least number to be added to 8200 to make it a perfect square?

a) 81 b) 100 c) 264 d) 154 e) None of the above

41. Find the difference in 777 and its nearest perfect square number.

a) 4 b) 7 c) 27 d) 28 e) None of the above

42. Find the value of $\sqrt{30+\sqrt{30+\ldots}}$

a) 5 b) $3\sqrt{10}$ c) 6 d) 7

43. Find the value of $\sqrt{9\sqrt{9\sqrt{9\ldots}}}$

a) 81 b) 3 c) 9 d) $\infty$

44. If $x = \sqrt{8+\sqrt{8+\sqrt{8\ldots}}}$ and $y = \sqrt{8-\sqrt{8-\sqrt{8\ldots}}}$, then

a) $x+y=1$ b) $x+y+1=0$ c) $x-y=1$ d) $x-y+1=0$

45. $\sqrt{6+\sqrt{6+\sqrt{6+\ldots}}}$ is equal to

a) 3 b) 2 c) 1.5 d) 2.5

46. Find the value of $\sqrt{6\sqrt{6\sqrt{6\sqrt{6\sqrt{6}}}}}$

a) 6 b) $6^{31/32}$ c) $6^{31/34}$ d) 3

47. If $\sqrt{24} = 4.899$, then find the value of $\sqrt{\frac{8}{3}}$

a) 1.633 b) 1.333 c) 2.666 d) 0.544

48. Find the value of $\sqrt[3]{2744} \div 7$

a) 14 b) 2 c) 15 d) 8 e) None of the above

49. Each student of class 10 contributed some money for a picnic. The money contributed by each student was equal to the cube of the total number of students. If the total collected amount was ₹29791, then find the total number of students.

a) 15 b) 27 c) 31 d) 34

50. (Smallest common multiple of 12 and 16) × (Smallest common multiple of 10 and 15) is equal to

a) 960 b) 720 c) 1440 d) 480

51. Find the value of $2 \times \sqrt{3}$ upto three places of decimal.

a) 1.732 b) 3.464 c) 4.464 d) 2.732 e) None of the above

52. If $\sqrt{2} = 1.4142$, then find the value of $2 \times \dfrac{\sqrt{2}}{(2+\sqrt{2})}$

a) 0.8284 b) 0.8184 c) 0.8283 d) 0.8483 e) None of the above

53. If $(\sqrt{a} + \sqrt{b}) = 17$ and $(\sqrt{a} - \sqrt{b} = 1)$, then find $\sqrt{ab}$

a) 17 b) 18 c) 72 d) 21 e) None of the above

54. If 46 is subtracted from the square of a number. The answer so obtained is 485. What is the number?

a) 49 b) 51 c) 56 d) 53 e) None of the above

**Exercise: Higher Skill Level Questions**

1. $\dfrac{1}{4} \times \sqrt{[(12.1)^2 - (8.1)^2] \div [(0.25)^2 + (0.25)(19,95)]}$

a) 8 b) 4 c) 1 d) 3 e) None of the above

2. $\dfrac{1}{\sqrt{9}-\sqrt{8}} - \dfrac{1}{\sqrt{8}-\sqrt{7}} + \dfrac{1}{\sqrt{7}-\sqrt{6}} - \dfrac{1}{\sqrt{6}-\sqrt{5}} + \dfrac{1}{\sqrt{5}-\sqrt{4}} = ?$

a) 0 b) 1 c) 1/3 d) 5 e) 1/5

3. If $3a = 4b = 6c$ and $a + b + c = 27\sqrt{29}$, then find $\sqrt{a^2 + b^2 + c^2}$

a) $3\sqrt{29}$ b) 81 c) 87 d) 89 e) None of the above

4. If $a = \dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$, then $a + \dfrac{1}{a}$ is equal to

a) 4 b) 6 c) 9 d) 10

5. If The expression $x + 809436 \times 809438$ be a perfect square, then the value of $x$ is

   a) 0 b) 1 c) 809436 d) 809438

6. The sum of two numbers when multiplied with each of the numbers separately, then the results of the above multiplications are 2418 and 3666, respectively. Find the difference between the numbers.

   a) 16 b) 21 c) 19 d) 23

7. A gardener has 1000 plants. He wants to plant them in such a way that the number of rows and the number of columns remains the same. What is the minimum number of plants that he needs more for this purpose?

   a) 14 b) 24 c) 32 d) 34

8. Find the value of

$$\sqrt{\frac{(0.08)^3 + (0.21)^2 + (0.066)^2}{(0.003)^2 + (0.021)^2 + (0.0066)^2}} + 2$$

   a) 2.8 b) 5 c) $10^2$ d) $5^2$ e) None of the above

9. $\sqrt[3]{1331} \times \sqrt[3]{216} \times \sqrt[3]{720} \times \sqrt[3]{64}$

   a) 13.62 b) 79 c) 14.82 d) 90.88 e) None of the above

10. A toy factory manufactured a batch of electronic toys. If the toys were packed in boxes of 115 each, 13 boxes would not be filled completely. If the toys were packed in boxes of 65 each, 22 such boxes would not be enough to pack all of them. Coincidentally, in the end, the toys were packed in $n$ boxes containing $n$ toys each, without any remainder. The total number of toys was

    a) 1424 b) 1434 c) 1444 d) 1454

11. The least number which is a perfect square and has 7936 as one of its factors is equal to

    a) 12.008 b) 246016 c) 61504 d) 240616 e) None of the above

12. The positive square root of $(\sqrt{48} - \sqrt{45})$

    a) $\frac{\sqrt[4]{3}}{\sqrt{2}}(\sqrt{5} - \sqrt{3})$ b) $\frac{\sqrt[4]{3}}{2}(\sqrt{5} - \sqrt{3})$ c) $\frac{\sqrt{2}}{\sqrt[4]{3}}(\sqrt{5} - \sqrt{3})$ d) $\frac{\sqrt{2}}{\sqrt[4]{3}}(\sqrt{5} + \sqrt{3})$

13. Find the value of $\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}\ldots}}}}}$

    a) 2 b) $2^2$ c) $2^3$ d) $2^5$

14. What should be added to the $x(x + a)(x + 2a)(x + 3a)$, so that the sum be a perfect square?

    a) $a^2$ b) $a^4$ c) $a^3$ d) None of the above

| Basic Skill Level Questions Answer Key | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | c | 6 | b | 11 | b | 16 | b | 21 | d | 26 | d | 31 | a | 36 | a | 41 | b | 46 | b | 51 | b |
| 2 | a | 7 | d | 12 | b | 17 | c | 22 | a | 27 | d | 32 | d | 37 | b | 42 | c | 47 | a | 52 | a |
| 3 | a | 8 | b | 13 | a | 18 | b | 23 | a | 28 | a | 33 | b | 38 | c | 43 | c | 48 | b | 53 | c |
| 4 | c | 9 | c | 14 | a | 19 | b | 24 | b | 29 | c | 34 | b | 39 | a | 44 | c | 49 | c | 54 | b |
| 5 | a | 10 | c | 15 | d | 20 | d | 25 | a | 30 | a | 35 | d | 40 | a | 45 | a | 50 | c | | |

| Higher Skill Level Questions | | | | | |
|---|---|---|---|---|---|
| 1 | c | 6 | a | 11 | b |
| 2 | d | 7 | b | 12 | a |
| 3 | c | 8 | b | 13 | a |
| 4 | d | 9 | b | 14 | b |
| 5 | b | 10 | c | | |

# Chapter 6- Simplification

1. $7059 - 2350 + 1936 = ? \times 50$

   a) 123.6 b) 132.3 c) 132.6 d) 123.9 e) None of the above

2. If $\sqrt{2} = 1.4142$ then the value of $\dfrac{7}{4+\sqrt{2}}$

   a) 3.5858 b) 4.4142 c) 1.2929 d) 1.5858

3. What is 786 times of 964?

   a) 759276 b) 749844 c) 75416 d) 757704 e) None of the above

4. Solve $\dfrac{(999+588)^2 - (999-588)^2}{999 \times 588}$

   a) 8 b) 3 c) 2 d) 4 e) None of the above

5. Solve $\dfrac{(238+131)^2 + (238-131)^2}{238 \times 238 + 131 \times 131}$

   a) 4 b) 2 c) 8 d) 9 e) None of the above

6. $\dfrac{\dfrac{1}{5} + 999 \times \dfrac{494}{495} \times 99}{4} = ?$

   a) 25000 b) 24225 c) 24800 d) 24750 e) None of the above

7. If $x = \sqrt{3} + \sqrt{2}$, then the value of $\left(x + \dfrac{1}{x}\right)$

   a) 2 b) 3 c) $2\sqrt{2}$ d) $2\sqrt{3}$

8. Find the value of $a^3 + b^3 + c^3 - 3abc$ when $a = 225,\ b = 226,\ c = 227$

   a) 2304 b) 2430 c) 2034 d) 2340

9. The value of $\dfrac{(0.96)^3 - (0.1)^3}{(0.96)^2 + 0.096 + 0.01}$ is

   a) 0.86 b) 1.06 c) 0.95 d) 0.97

10. Value of $\dfrac{\sqrt{0.0001} \times \sqrt[3]{1000000}}{\sqrt[3]{0.125} \times \sqrt[3]{0.0008}}$

    a) 1 b) 10 c) 0.1 d) 0.01

11. $3.5 \times (60 \div 0.5) = ?$

    a) 62 b) 96 c) 74 d) 88 e) None of the above

12. Simplify $6 - [9 - \{18 - (15 - \overline{12 - 9})\}]$

a) 1 b) 4 c) 5 d) 3 e) None of the above

13. $3\frac{2}{3} + 2\frac{3}{4} + 1\frac{1}{2} = ?$

a) $8\frac{11}{12}$ b) $10\frac{12}{13}$ c) $7\frac{11}{12}$ d) $9\frac{11}{13}$ e) None of the above

14. $\frac{5}{8} \times 5\frac{3}{5} \div \frac{4}{9} = ?$

a) $2\frac{13}{27}$ b) $1\frac{11}{27}$ c) $2\frac{23}{32}$ d) $3\frac{21}{32}$ e) None of the above

15. $10.8 \times 5.5 \times 84 = ?$

a) 458.69 b) 489.96 c) 498.96 d) 485.69 e) None of the above

16. $(4 \times 4 \times 4 \times 4 \times 4 \times 4)^5 \times (4 \times 4 \times 4)^8 + (4)^3 = (64)^{?}$

a) 17 b) 10 c) 16 d) 11 e) None of the above

17. $8648 - 7652 = ? \times 40$

a) 24.7 b) 28.9 c) 21.2 d) 25.9 e) None of the above

18. $683.46 - 227.39 - 341.85 = ?$

a) 114.22 b) 141.22 c) 144.22 d) 112.22 e) None of the above

19. $\frac{13}{63} \div \frac{104}{14} \times \frac{52}{19} = ?$

a) $\frac{12}{273}$ b) $\frac{13}{171}$ c) $\frac{17}{171}$ d) $\frac{18}{171}$ e) None of the above

20. $(5568 \div 87)^{\frac{1}{3}} + (72 \times 2)^{\frac{1}{2}} = (?)^{\frac{1}{2}}$

a) 256 b) 4 c) $\sqrt{2}$ d) 16 e) None of the above

21. $\sqrt{13^2 + 28 + 4 - (3)^2 + 107} = (?)^2$

a) 2 b) 16 c) 256 d) 4 e) $256^2$

22. $(0.49)^4 \times (0.343)^4 \div (0.2401)^4 = (70 \div 100)^{?+3}$

a) 3 b) 1 c) 4 d) 7 e) None of the above

23. $\dfrac{5 - \left[\frac{3}{4} + \left\{2\frac{1}{2} - \left(\frac{1}{2} + \overline{\frac{1}{6} - \frac{1}{7}}\right)\right\}\right]}{2} = ?$

a) $1\frac{23}{168}$ b) $2\frac{23}{168}$ c) $3\frac{23}{168}$ d) $4\frac{23}{168}$ e) None of the above

24. Simplify $\dfrac{\left[3\frac{1}{4}+\left\{1\frac{1}{4}-0.5\left(2\frac{1}{2}-\overline{\frac{1}{4}-\frac{1}{6}}\right)\right\}\right]}{4\times\frac{1}{12}}$

a) 245 b) 233 c) 234 d) 299 e) None of the above

25. $\dfrac{0.5\times0.5\times0.5+0.2\times0.2\times0.2+0.3\times0.3\times0.3-3\times0.5\times0.3\times0.2}{0.5\times0.5+0.2\times0.2+0.3\times0.3-0.5\times0.2-0.2\times0.3-0.5\times0.3}=?$

a) 1 b) 0.6 c) 0.4 d) 0.03

26. $\dfrac{(2.247)^3+(1.730)^3+(1.023)^3-3\times2.247\times1.730\times1.023}{(2.247)^2+(1.730)^2+(1.023)^2-2.247\times1.730-1.730\times1.023-2.247\times1.023}=?$

a) 1.730 b) 4 c) 5 d) 5.247

27. What is the value of $\dfrac{725\times725\times725+371\times371\times371}{725\times725-725\times371+371\times371}=?$

a) 9610 b) 1960 c) 1096 d) 1016

28. $\dfrac{8.73\times8.73\times8.73+4.27\times4.27\times4.27}{8,73\times8.73-8.73\times4.27+4.27\times4.27}$ is

a) 11 b) 13 c) 11/7 d) None of the above

29. $\sqrt{289}-\sqrt{625}+\sqrt{25}$ is equal to

a) 17 b) 15 c) 12 d) -8/5 e) None of the above

30. $a^2+b^2=234$ and $ab=108$, find the value of $\dfrac{a+b}{a-b}$

a) 10 b) 8 c) 5 d) 4 e) None of the above

31. If $a^2+1=a$, then the value of $a^{12}+a^6+1$ is

a) 2 b) 3 c) -3 d) 1

32. If $x=0.764,\ y=1.236$, then what is the value of $\left(\dfrac{y-x}{y+x}\right)$?

a) 0.764 b) 0.236 c) 2 d) 0.472

33. If $a, b$ and $c$ are non-zero, $a+\dfrac{1}{b}=1$ and $b+\dfrac{1}{c}=1$, then the value of $abc$ is

a) -3 b) 1 c) -1 d) 3

34. If $a+b+c=0$, then the value of $\left(\dfrac{a+b}{c}+\dfrac{b+c}{a}+\dfrac{c+a}{b}\right)\left(\dfrac{a}{b+c}+\dfrac{b}{c+a}+\dfrac{c}{a+b}\right)$is

a) 9 b) 0 c) 8 d) -3

35. If $a * b = a + b + \frac{a}{b}$, then the value of $12 * 4$ is

a) 48 b) 19 c) 20 d) 21

36. If $x = \frac{\sqrt{2}+1}{\sqrt{2}-1}$ and $x - y = 4\sqrt{2}$, then the value of $(x^2 + y^2)$ is

a) 34 b) 38 c) 30 d) 32

37. If $x + \frac{a}{x} = 1$, then the value of $\frac{x^2 + x + a}{x^3 - x^2}$ is

a) -2 b) $-\frac{a}{2}$ c) $\frac{2}{a}$ d) $-\frac{2}{a}$

38. If $\sqrt{28 - 6\sqrt{3}} = \sqrt{3}a + b$, (where $a, b$ are rationals), value of $(a + b)$ is

a) -2 b) 2 c) 1 d) -1

39. If $x + y = 1$, then the value of $(x^3 + y^3 + 3xy)$ is

a) -2 b) 1 c) 2 d) 3

40. If $a = (\sqrt{2} - 1)^{1/3}$, then the value of $(a - a^{-1})^3 + 3(a - a^{-1})$ is

a) -2 b) 2 c) $2\sqrt{2}$ d) $\sqrt{2}$

41. If $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ and $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$, then the value of $\frac{x^2}{y} + \frac{y^2}{x}$ is

a) 52 b) 76 c) 4 d) 64

42. If $p + q = 10$ and $pq = 5$, then the numerical value of $\frac{p}{q} + \frac{q}{p}$ will be

a) 22 b) 18 c) 16 d) 20

43. If $x + y = 18$ and $xy = 72$, what is the value of $x^2 + y^2$?

a) 120 b) 90 c) 180 d) Cannot be determined e) None of the above

44. $\frac{(m-n)^3 + (n-r)^3 + (r-m)^3}{6(m-n)(n-r)(r-m)} = ?$

a) $\frac{1}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{5}$ d) $\frac{1}{6}$ e) None of the above

45. If $a + b + c = 14$ and $a^2 + b^2 + c^2 = 96$, then $(ab + bc + ca) = ?$

a) 51 b) 55 c) 50 d) 65 e) None of the above

46. $\frac{1}{1\times4} + \frac{1}{4\times7} + \frac{1}{7\times10} + \frac{1}{10\times13} + \frac{1}{13\times16} = ?$

a) $\frac{5}{16}$ b) $\frac{3}{16}$ c) $\frac{7}{16}$ d) $\frac{11}{16}$

47. $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{m}\right)$ =?

a) $\frac{1}{m}$ b) $m$ c) $m+1$ d) $\frac{1}{(m-1)}$ e) None of the above

48. $\left(2-\frac{1}{3}\right)\left(2-\frac{3}{5}\right)\left(2-\frac{5}{7}\right)\ldots\left(2-\frac{997}{999}\right)$ is equal t

a) $\frac{1001}{999}$ b) $\frac{999}{1001}$ c) $\frac{1001}{3}$ d) $\frac{5}{1001}$

49. The value of $\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\ldots\left(1+\frac{1}{150}\right)$ is

a) 65.5 b) 50.5 c) 105 d) 75.5 e) None of the above

50. If $x=\frac{\sqrt{3}}{2}$, then the value of $\frac{1+x}{1+\sqrt{1+x}}+\frac{1-x}{1-\sqrt{1-x}}$ is equal to

a) 0 b) 1 c) $\frac{\sqrt{3}}{2}$ d) $\sqrt{3}$

51. If $a+\frac{1}{a}=\sqrt{3}$, then the value of $a^6-\frac{1}{a^6}+2$ will be

a) $3\sqrt{3}$ b) 5 c) 1 d) 2

52. If $x+\frac{1}{x}=2$, then $\frac{2x^2+2}{3x^2+5x+3}$ =?

a) $\frac{4}{11}$ b) $\frac{1}{2}$ c) $1\frac{3}{4}$ d) $2\frac{1}{3}$

53. If $a+\frac{1}{b}=1$ and $b+\frac{1}{c}=1$, then $c+\frac{1}{a}$ =?

a) 4 b) 24 c) 3 d) 1

54. If $x+\frac{a}{x}=b$, then value of $\frac{x^2+bx+a}{bx^2-x^3}$ is

a) $a+b$ b) $\frac{2b}{a}$ c) $\frac{b}{a}$ d) $ab$

55. If $x+\frac{1}{x}=6$, then $x^4+\frac{1}{x^4}$ is

a) 1152 b) 1154 c) 1148 d) 1150

56. If $x+\frac{1}{x}=2$, then what is value of $x-\frac{1}{x}$?

a) 0 b) 1 c) 2 d) -2

57. If $x + \frac{1}{x} = 3$, then $x^5 + \frac{1}{x^5}$ is equal to

a) 123 b) 83 c) 92 d) 112

58. $3x + 2y = 12$ and $xy = 6$, then the value of $9x^2 + 4y^2$ is

a) 76 b) 80 c) 72 d) 74 e) None of the above

59. If $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$, then what is $\left(x^3 - \frac{1}{x^3}\right)$ equal to ?

a) $\frac{75}{16}$ b) $\frac{63}{8}$ c) $\frac{95}{8}$ d) None of the above

**Exercise : Higher Skill Level Questions**

1. A 120 m long rope is cut into 3 equal parts. What is the length of each part?

a) 40 m b) 30 m c) 50 m d) 25 m e) None of the above

2. When a number is multiplied 5 times with itself, it gives the value 1445. Find the number.

a) 18 b) 15 c) 19 d) 17 e) None of the above

3. A board 80 inches long is divided into 4 equal parts. What is the length of each part?

a) 20 inches b) 25 inches c) 15 inches d) 18 inches e) None of the above

4. A man spends 2/5 of his income on house rent, 3/10 on food and 1/8 on conveyance. If he has ₹2800 left with him, find his expenditure on food and conveyance together.

a) ₹7000 b) ₹5500 c) ₹6700 d) ₹6800 e) None of the above

5. Mani Ram divides ₹17200 amongst his 5 sons, 4 daughters and 2 friends. If each daughter receives four times as much as each friend receives and each son receives five times as much as each friend receives, how much does each daughter receive?

a) ₹800 b) ₹1500 c) ₹1000 d) ₹1600

6. One-third of Rahul's marks in Mathematics exceeds one-half of his marks in Hindi by 30. If he got 480 marks in the two subjects together, how many marks did he get in Hindi?

a) 200 b) 294 c) 156 d) 196 e) None of the above

7. A container of milk was 4/5 full. When 12 bottles of milk were taken out and 8 bottles of milk were poured into it, it was 3/4 full. How many bottles of milk can the container contain?

a) 80 b) 40 c) 90 d) 30 e) None of the above

8. The highest score in an innings was $\frac{3}{11}$ of the total score and the next highest was $\frac{3}{11}$ of the remainder. If the scores differ by 18, find the total score

a) 244 b) 242 c) 294 d) 255 e) None of the above

9. Mohan gets 3 marks for each correct sum and loses 2 marks for each wrong sum. He attempts 30 sums and obtains 40 marks. The number of sums solved correctly is

    a) 25 b) 10 c) 15 d) 20

10. The number obtained by interchanging the digits of a two-digit number is more than the original number by 27 and the sum of the digits is 13. What is the original number?

    a) 58 b) 67 c) 76 d) 85

11. Of three positive numbers the product of the first and second is 42, that of the second and third is 56 and that of third and first is 48. The third number is

    a) 12 b) 6 c) 7 d) 8

12. It is required to get 40% marks to qualify an exam. A candidate scored 200 marks and failed by 8 marks. What were the maximum marks of that exam?

    a) 520 b) 540 c) 502 d) Couldn't be determined e) None of the above

13. Mr. Arun is on tour and he has ₹360 for his expenses. If he exceeds his tour by 4 days, he must cut down his daily expenses by ₹3. For how many days is Mr. Arun out on tour?

    a) 40 b) 20 c) 60 d) 15 e) None of the above

14. From a group of men and women, 15 women leave from it. Then, there are 2 men for each women. After this, 45 men leave. Then, there are 5 women for each men. Find the double of the number of women in the beginning.

    a) 25 b) 23 c) 35 d) 80 e) None of the above

15. Bus fare between Raipur and Mirpur for one adult is six times the fare for one child. If an adult's bus fare is ₹114, how much amount will be paid by 4 adults and 5 children together for travelling the same distance?

    a) ₹505 b) ₹551 c) ₹572 d) ₹560 e) None of the above

16. The total cost of 12 pens and 5 pencils is equal to ₹111. Also, the cost of 1 pencil is ₹5 less than the cost of 1 pen. What will be the cost of 8 pens and 9 pencils?

    a) ₹89 b) ₹97 c) ₹91 d) ₹78 e) None of the above

17. When an amount is distributed amongst 14 girls, each of them gets ₹160 more than the amount received by each girl in that condition when the same amount is distributed equally amongst 18 girls. Find the amount.

    a) ₹5040 b) ₹10070 c) ₹10080 d) ₹5000 e) None of the above

18. A man had 170 currency notes in all, some of which were of ₹100 denominations and some of ₹50 denominations. The total amount of all these currency notes was ₹10000. How much amount did he have in the denominations of ₹50?

    a) ₹4000 b) ₹9000 c) ₹7000 d) ₹6000 e) None of the above

19. Mr. Sahni employed a worker for a certain work to be done in some days. He pays ₹20 to the worker for each working day and the payment is reduced by ₹3 for each non-working day of the worker. For how many days the worker does remain absent from work if at the end of 120 days, he gets ₹560 as total remuneration?

    a) 80 b) 60 c) 70 d) 40 e) None of the above

20. A fruit seller has a crate of apples containing one bruised apple for every 40 apples in the crate. If 3 out of every 4 bruised apples are considered unsaleable and there are 9 unsaleable apples in the crate, then how many apples are there in the crate?

    a) 500 b) 390 c) 632 d) 480 e) None of the above

21. City A has a population of 136000 which is decreasing at the rate of 2400 per year. City B has a population of 84000 which is increasing at the rate of 1600 per year. In how many years will the population of two cities become equal?

    a) 15 b) 18 c) 13 d) 19 e) None of the above

**Basic Skill Level Questions Answer Key**

| | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | e | 6 | d | 11 | e | 16 | a | 21 | d | 26 | c | 31 | b | 36 | a | 41 | a | 46 | a | 51 | d | 56 | a |
| 2 | c | 7 | d | 12 | d | 17 | e | 22 | b | 27 | c | 32 | b | 37 | d | 42 | b | 47 | a | 52 | a | 57 | a |
| 3 | d | 8 | c | 13 | c | 18 | a | 23 | a | 28 | b | 33 | c | 38 | a | 43 | c | 48 | c | 53 | d | 58 | c |
| 4 | d | 9 | a | 14 | d | 19 | b | 24 | c | 29 | c | 34 | a | 39 | b | 44 | a | 49 | d | 54 | b | 59 | b |
| 5 | b | 10 | b | 15 | c | 20 | a | 25 | a | 30 | c | 35 | b | 40 | a | 45 | c | 50 | b | 55 | b | | |

**Higher Skill Level Questions**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | a | 6 | c | 11 | d | 16 | c | 21 | c |
| 2 | d | 7 | a | 12 | a | 17 | c | | |
| 3 | a | 8 | b | 13 | b | 18 | c | | |
| 4 | d | 9 | d | 14 | d | 19 | a | | |
| 5 | d | 10 | a | 15 | b | 20 | d | | |